

Test report based on BSI AIS 20 / AIS 31

2026-Apr-29 16:01:29.029021

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Identification information

1.1 Identification information of input data

- Filename of input data : See Annex A.1.
- Name of the submitter of the input data :
- Brief explanation of the input data :

1.2 Identification of analysis environment

Table 1 Identification information of analysis environment

Analysis tool	Name	Another AIS 20/AIS 31 test tool
	Versioning information	3.0.6
	built as	64-bit application
	built by	Visual Studio 2019 version 16.11 (_MSC_FULL_VER: 192930159)
	linked libraries	Boost C++ 1.91.0
	with	OpenMP disabled
Analysis environment	Hostname	██████████
	CPU information	Intel(R) Core(TM) i5-██████████
	Physical memory size	██████ MiB
	OS name	Microsoft Windows 11 Pro
	OS version	10.0.26200 N/A Build 26200
	System type	64-bit
	Username	██████

1.3 Identification of analysis conditions

Table 2 Identification information of analysis conditions

Bits per sample	8
Byte to bit conversion	Most Significant bit (MSb) first

1.4 Identification of analysis method

Black Box Test Suite T_{irr_n} is applied, based on BSI AIS 20 [1] / AIS 31[2] and mathematical-technical appendix [3], together with corrections [4].

2

Executive summary

2.1

Test results based on BSI AIS 20 / AIS 31

Table 3 Test results

Tests	1-st trial		2-nd trial	
	Pass / Fail	Notes	Pass / Fail	Notes
Test T1 (monobit test)	Pass	see 3.1.1	N/A	—
Test T2 (poker test)	Pass	see 3.2.1	N/A	—
Test T3 (MultiMMC Prediction Estimate)	Pass	see 3.3.1	N/A	—
Test T4 (LZ78Y Prediction Estimate)	Pass	see 3.4.1	N/A	—
Overall test result	Pass			

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Detailed information of tests for given input data

3.1

Test T1 (monobit test)

3.1.1

Supplemental information for traceability for the 1-st trial

Table 4 Supplemental information for traceability (Test T1) for the 1-st trial

Symbol	Value
$t_{IRN(1)}$	9994

3.2

Test T2 (poker test)

3.2.1

Supplemental information for traceability for the 1-st trial

Table 5 Supplemental information for traceability (Test T2) for the 1-st trial

Symbol	Value
$t_{IRN(2)}$	14.7392

3.3 The MultiMMC Prediction Estimate (Test T3)

3.3.1 Distribution of *correct* for the 1-st trial

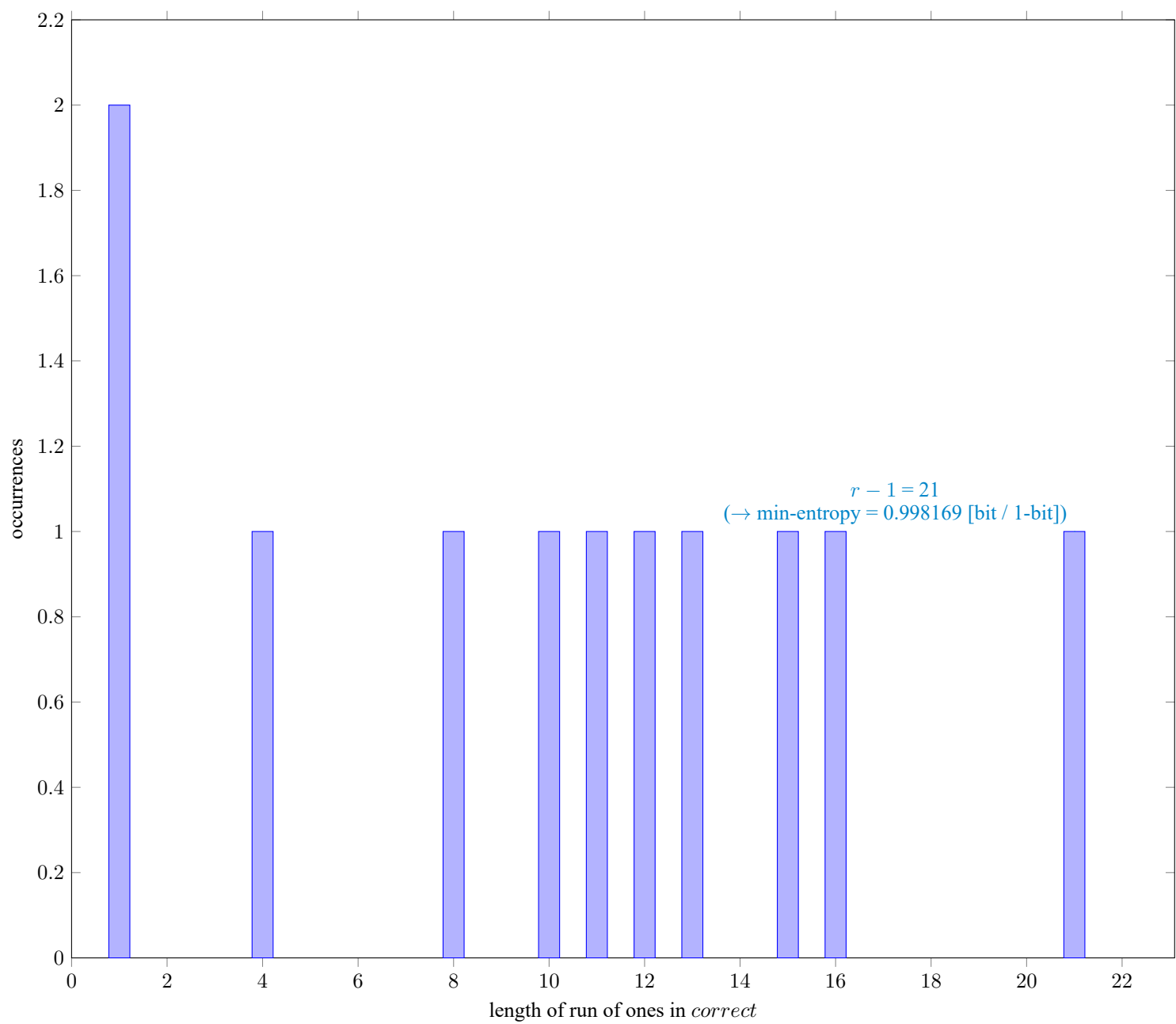


Fig. 1 Distribution of *correct*

3.3.2 Supplemental information for traceability for the 1-st trial

Table 6 Supplemental information for traceability (NIST SP 800-90B Section 6.3.9) for the 1-st trial

Symbol	Value
N	999998
C	499346
P_{global}	0.499347
P'_{global}	0.500635
r	22
P_{local}	0.444709

3.4 The LZ78Y Prediction Estimate (Test T4)

3.4.1 Distribution of *correct* for the 1-st trial

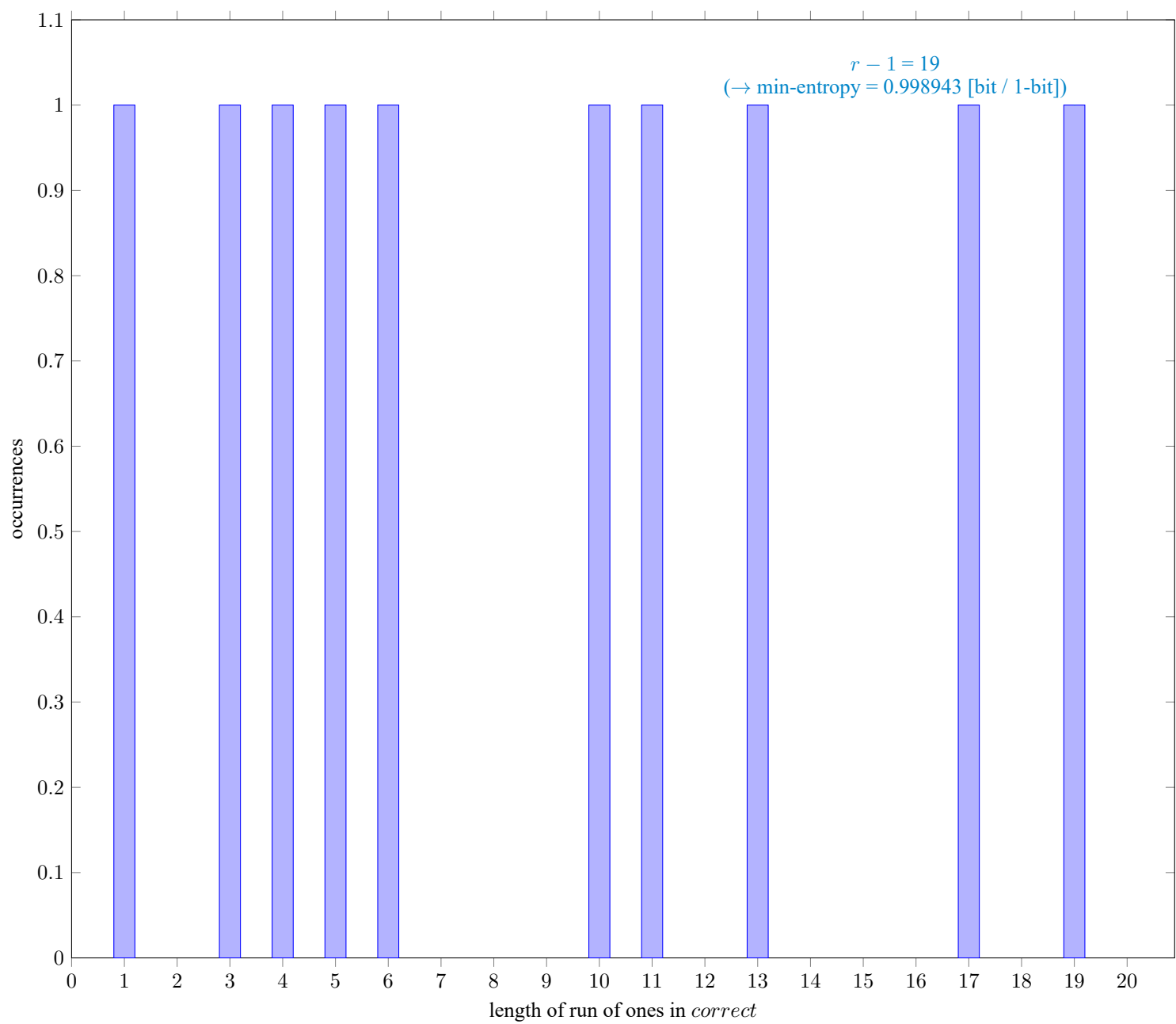


Fig. 2 Distribution of *correct*

3.4.2 Supplemental information for traceability for the 1-st trial

Table 7 Supplemental information for traceability (NIST SP 800-90B Section 6.3.10) for the 1-st trial

Symbol	Value
N	999983
C	499070
P_{global}	0.499078
P'_{global}	0.500366
r	20
P_{local}	0.408812

A

Identification information

A.1

Identification of input data

Table 8: Identification information of input data

No	Item	Value
1	URL of the input data for Tests T1 through T4	https://github.com/g-g-sakura/AnotherAIS31TestTool/tree/main/sample_input_data/v3_sample_input_data_for_T1-T4.bin
	SHA-256 hash value of the input data for Tests T1 through T4[hex]	a3b04689 c5512355 ba82b434 ea389f77 fd2883e2 b49d6c1b 479788b3 622dfdc1
	Last write time	2024-Nov-17 11:22:52
		end of the table

B References

- [1] Bundesamt für Sicherheit in der Informationstechnik (BSI), *Anwendungshinweise und Interpretationen zum Schema (AIS) - Funktionalitätsklassen und Evaluationsmethodologie für deterministische Zufallszahlengeneratoren*, Version 4.0 (11.04.2025), https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/Zertifizierung/Interpretationen/AIS_20_pdf.pdf?__blob=publicationFile&v=4
- [2] Bundesamt für Sicherheit in der Informationstechnik (BSI), *Anwendungshinweise und Interpretationen zum Schema (AIS) - Funktionalitätsklassen und Evaluationsmethodologie für physikalische Zufallszahlengeneratoren*, Version 4.0 (11.04.2025), https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/Zertifizierung/Interpretationen/AIS_31_pdf.pdf?__blob=publicationFile&v=5
- [3] Matthias Peter and Werner Schindler, *A Proposal for Functionality Classes for Random Number Generators*, Version 3.0 (September 10, 2024), https://www.bsi.bund.de/SharedDocs/Downloads/EN/BSI/Certification/Interpretations/AIS_31_Functionality_classes_for_random_number_generators_e_2024.pdf?__blob=publicationFile&v=3
- [4] G. Sakurai, *Proposed list of corrections for NIST SP 800-90B 6.3 Estimators*, June 2025 https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/ProposedListOfCorrections_SP800-90B.pdf